

SDLC AutoPilot AI: Agentic Automation of Software Development Life Cycle

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Abstract— The Software Development Life Cycle (SDLC) remains one of the most resource-intensive processes in modern engineering, requiring continuous coordination across requirement gathering, planning, team formation, testing, and deployment. Despite the availability of advanced automation tools, much of this process still depends on manual oversight, which often results in delays, misaligned requirements, and costly rework. To address these challenges, we propose SDLC AutoPilot AI, an intelligent agentic system designed to orchestrate the entire SDLC in an end-to-end manner. Beginning from a high-level project brief, the system leverages large language models and retrieval-augmented pipelines to transform unstructured inputs into well-structured requirements. It then prepares project plans and budget estimates, quantifying uncertainty ranges to improve transparency in forecasting. Using automated reasoning and supervised models, the system performs role assignment, forms collaborative teams within enterprise platforms, and provisions the necessary infrastructure for execution. Furthermore, SDLC AutoPilot AI integrates tightly with collaboration tools, version control platforms, and CI/CD pipelines to support automated verification, test case generation, and staged deployments. Preview environments are provisioned dynamically, enabling early client validation and reducing downstream rework. To ensure accountability and resilience, the system includes governance guardrails for human approval, observability features such as decision logs and audit trails, and safeguards against hallucination and misclassification. This paper presents the architectural design of the system, details the algorithms used for estimation and role allocation, and outlines the integration pathways with enterprise ecosystems. An evaluation framework is also proposed to assess its effectiveness in terms of delivery speed, reduction in rework, defect density, and stakeholder satisfaction. Finally, we reflect on the broader implications of introducing agentic automation into software engineering workflows, with particular attention to governance, reliability, and ethical considerations.

Keywords— *Software Development Life Cycle (SDLC), Agentic AI, Large Language Models (LLMs), Project Automation, Software Engineering, Intelligent Orchestration, Requirements Engineering, Continuous Integration (CI), Continuous Deployment (CD), Governance, Observability, AI in Software Development, Delay Prediction, Role Assignment, Workflow Automation.*

I. INTRODUCTION

Software development is a **highly collaborative and complex process** requiring continuous coordination among multiple stakeholders and tools. The need to coordinate among product managers, developers, testers, and clients often makes it the **most challenging part of the SDLC**. Poor coordination, such as misaligned requirements or delayed feedback, creates bottlenecks that lead to **rework, missed deadlines, and increased delivery costs**.

One of the central challenges in software engineering is **requirements management**. Requirements are often

ambiguous, incomplete, or constantly evolving. If requirements are not clearly defined at the start, downstream teams may misinterpret them, leading to miscommunication and wasted effort. Client or stakeholder feedback often arrives only after substantial work is complete, at which point changes are costly and can require re-engineering entire modules. This phenomenon, known as **scope churn**, is a major driver of project delays and budget overruns.

Another pain point emerges from **toolchain fragmentation**. Modern development relies on a diverse ecosystem of tools—project trackers, version control systems, CI/CD pipelines, communication platforms, and testing frameworks. While each of these tools is powerful in isolation, they are often poorly integrated. Developers spend a significant amount of time switching between environments, duplicating data across tools, and managing dependencies manually. The lack of seamless orchestration reduces productivity and increases the risk of errors, especially in large distributed teams.

Over the past few years, automation has been introduced into several parts of the SDLC to mitigate these challenges. Continuous integration pipelines help enforce quality checks, agile tools track progress, and cloud platforms provide scalable deployment environments. However, most of these automation efforts are **narrow in scope**. They target specific tasks—such as running unit tests or deploying containers—rather than coordinating the entire workflow. The burden of decision-making, planning, and cross-tool synchronization still rests on human project managers and engineering leads. This leaves significant room for inefficiency.

Recent advances in **Large Language Models (LLMs) and autonomous AI agents** have opened new possibilities for addressing these inefficiencies. Unlike traditional rule-based automation, LLM-driven systems can interpret unstructured project briefs, reason over complex dependencies, and adapt dynamically to changes in context. Combined with structured orchestration frameworks, they can act not only as assistants for individual developers but also as *coordinators* that manage tasks across the full SDLC. This shift represents a new paradigm: moving from task-level automation to **workflow-level automation**.

In this context, we introduce **SDLC AutoPilot AI**, an intelligent agentic system designed to manage the end-to-end software development process. The system builds on the recent progress in LLMs, retrieval-augmented pipelines, and orchestration platforms to address some of the most persistent challenges in software engineering. Its goal is not to replace human judgment, but to reduce repetitive effort, increase alignment among stakeholders, and improve delivery outcomes.

At its core, SDLC AutoPilot AI automates several key activities:

1. **Requirement Gathering** – The system parses project briefs, documents, and multi-turn conversations to generate structured Software Requirement Specifications (SRS). By leveraging natural language understanding and entity extraction, it produces requirements that are both precise and testable.
2. **Project Planning and Budgeting** – Using hybrid estimation techniques, the system prepares work breakdown structures, identifies dependencies, and predicts effort requirements. Unlike traditional estimation models that rely only on parametric formulas, AutoPilot integrates machine learning to calibrate predictions based on real project data.
3. **Team Assignment** – The system performs intelligent role allocation by matching tasks with available resources. Fairness constraints ensure balanced workload distribution, while skill-rotation features help teams grow capabilities over time.
4. **Feedback through Preview Environments** – For every new feature, AutoPilot provisions ephemeral preview environments. Clients and stakeholders can interact with these environments, provide structured feedback, and request changes before final integration. This reduces rework and ensures alignment between delivered features and client expectations.
5. **Automated Testing and Verification** – Derived from the requirements, AutoPilot generates unit, integration, and end-to-end test cases. It integrates these with CI/CD pipelines, sets up quality gates, and isolates flaky tests to maintain reliability.
6. **Deployment Management** – The system manages staged rollouts, ensuring that deployments progress smoothly from testing to production while incorporating human approvals at critical points.

By orchestrating these activities under a unified framework, SDLC AutoPilot AI addresses the long-standing problem of coordination gaps in software projects. Instead of developers and managers manually moving tasks between systems, the AI agents act as mediators that keep the process flowing. Importantly, human oversight remains a central principle of the design. Critical decision points—such as requirement baselines and production deployments—still require explicit approval from stakeholders. This ensures that automation enhances rather than replaces human accountability.

The broader implications of adopting such a system extend beyond just efficiency gains. By reducing rework and shortening feedback cycles, SDLC AutoPilot AI enables organizations to deliver higher-quality software faster. It also creates opportunities for more inclusive participation, as non-technical stakeholders can interact directly with preview environments and structured feedback channels. Moreover, the system's reliance on observability—through decision logs and audit trails—provides transparency that is often missing in conventional project management.

Nevertheless, the deployment of agentic AI in software development also raises important considerations around governance, reliability, and ethics. Risks such as model hallucination, biased decision-making, or estimator miscalibration must be carefully managed through guardrails and ongoing monitoring. Ensuring that automation complements rather than undermines human expertise remains a central challenge.

In summary, the growing complexity of software projects demands new approaches to coordination and automation. SDLC AutoPilot AI represents a step toward holistic workflow management powered by intelligent agents. By automating requirements gathering, planning, assignment, testing, and deployment—while embedding governance and human oversight—the system aims to reduce delays, improve alignment, and enhance overall project outcomes. This paper presents the architecture, algorithms, and evaluation framework of the system, and explores its potential impact on the practice of software engineering.

II. LITERATURE REVIEW

Boehm's *Software Engineering Economics* [1] laid the foundation for cost estimation in software projects, introducing models such as COCOMO that remain influential decades later. These parametric approaches, while effective in structured environments, often struggle with uncertainty and evolving requirements, motivating the search for more adaptive methods.

Building on this foundation, Jørgensen and Shepperd [3] conducted a systematic review of software cost estimation studies, highlighting the variability of prediction accuracy across contexts and emphasizing the need for hybrid models that combine historical data with contextual project features. Their work directly informs modern approaches that integrate statistical baselines with machine learning residual predictors.

The **SWEBOK Guide** [4] further codified knowledge areas across the SDLC, offering a standard reference framework for practices ranging from requirements engineering to testing. By outlining these processes formally, SWEBOK provides the structural backbone against which automation systems such as SDLC AutoPilot AI can be evaluated.

In recent years, the rise of **machine learning for code and software artifacts** has opened new possibilities for automation. Allamanis et al. [8] surveyed techniques for mining big code, applying probabilistic and deep learning models to tasks like code completion, bug detection, and refactoring. Their insights demonstrate how statistical models can capture software's inherent regularities, paving the way for LLM-powered assistants.

The arrival of **large language models (LLMs)** has further accelerated this trend. OpenAI's GPT-4 Technical Report [9] documents the capabilities of LLMs in understanding, generating, and reasoning over natural language and code, highlighting their potential role as multi-purpose assistants across the SDLC. Similarly, Chen et al. [11] introduced *Codex*, a model fine-tuned on source code, and evaluated its effectiveness in generating functional programs, demonstrating that LLMs can move beyond natural language processing into software engineering tasks.

Alongside these AI advancements, the software engineering community has developed new paradigms for continuous delivery and collaboration. Ebert et al. [15] provide a comprehensive overview of **DevOps**, detailing how

the integration of development and operations shortens feedback loops and improves software quality. Their work underscores the importance of toolchain integration and automation—key goals shared by SDLC AutoPilot AI.

Finally, Brodie’s early contributions on **agent-oriented software engineering** [19] laid the conceptual foundation for treating autonomous agents as first-class entities in the software process. This perspective resonates strongly with today’s movement toward agentic AI systems, where intelligent agents can coordinate tasks, communicate, and adapt within complex workflows.

III. BACKGROUND

Software engineering has long relied on structured frameworks and estimation models to manage complexity in large-scale projects. One of the earliest and most influential contributions in this area is Boehm’s *Software Engineering Economics* [1], which introduced the COCOMO family of cost estimation models. These models provided a parametric basis for predicting effort and schedule based on software size, becoming a standard reference point for project planning. Subsequent studies, such as the systematic review by Jørgensen and Shepperd [3], highlighted both the strengths and limitations of such approaches, emphasizing the variability of accuracy across project contexts and motivating the development of hybrid techniques that integrate historical data with adaptive predictors.

To unify the diverse practices within the field, the IEEE *Software Engineering Body of Knowledge (SWEBOK)* [4] codified key activities across the SDLC, including requirements engineering, project planning, testing, and maintenance. This framework serves as an authoritative guide, defining the very processes that automation systems such as SDLC AutoPilot AI aim to orchestrate.

Parallel to these methodological foundations, advances in artificial intelligence have increasingly shaped software engineering. The survey by Allamanis et al. [8] demonstrated how statistical and machine learning techniques could be applied to “big code” for tasks like code completion, defect detection, and automated refactoring. These approaches revealed the latent regularities present in software repositories and laid the groundwork for more sophisticated AI-driven tools.

The emergence of **large language models (LLMs)** has pushed these boundaries further. OpenAI’s GPT-4 Technical Report [9] showcased the ability of LLMs to understand and generate both natural language and code, making them suitable for multi-purpose roles in the SDLC. Complementing this, Chen et al. [11] introduced *Codex*, a code-specialized LLM, and demonstrated its capacity to generate functional programs from natural language prompts. Together, these developments mark a transition from narrow task automation to versatile AI systems capable of contributing across multiple stages of software development.

At the same time, organizational and infrastructural innovations have redefined how software is delivered. The rise of **DevOps**, described comprehensively by Ebert et al. [15], emphasized the integration of development and operations, enabling shorter feedback loops, continuous delivery, and closer alignment between stakeholders. This paradigm underscores the importance of toolchain integration—an aspect that is central to the design of SDLC AutoPilot AI.

Finally, the conceptual roots of using intelligent agents in software processes can be traced back to Brodie’s work on **agent-oriented software engineering** [19]. By framing

agents as autonomous, communicative entities capable of managing tasks and interactions, this perspective anticipated the current wave of agentic AI systems. In today’s context, such agentic approaches are particularly relevant as they enable distributed orchestration of complex workflows within software engineering.

Together, these foundational works—spanning cost estimation, process frameworks, machine learning, large language models, DevOps, and agent-oriented paradigms—establish the intellectual landscape on which SDLC AutoPilot AI is built.

IV. System Overview

A. Problem Formulation

When a new project brief **B** is received, along with a resource pool **R** (covering people, skills, and availability) and a set of constraints **C** (such as budget and timeline), the system’s objective is to generate the following outputs:

- 1) **Plan (P)**: A work breakdown structure (WBS), project schedule, and budget, including quantified uncertainty.
- 2) **Assignment (A)**: A complete team roster with defined roles and responsibilities.
- 3) **Milestone Graph (M)**: A task-level representation modeled as state machines, enriched with telemetry for tracking progress.

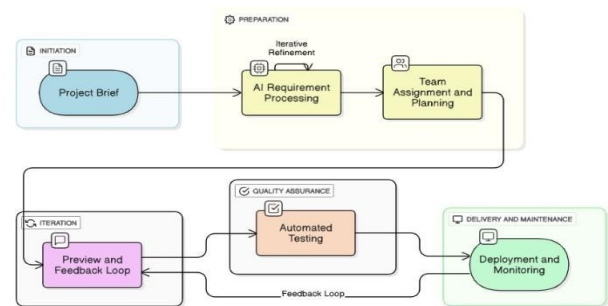


Figure. 1. SDLC AutoPilot AI: agent interactions and data flow.

B. Architecture (Agents)

Figure 1 illustrates the high-level architecture of SDLC AutoPilot AI. The system is composed of several specialized agents, each responsible for a key part of the workflow:

- 1) **Requirement Agent**: Parses project briefs, extracts entities and acceptance criteria, and maps them into a structured software requirements specification (SRS).
- 2) **Planning & Estimation Agent**: Builds the work breakdown structure (WBS) and produces P50/P80 effort and cost estimates using a hybrid method that combines parametric baselines with a learned residual model.
- 3) **Role Assignment Agent**: Matches tasks to resources by considering skills, availability, and project constraints.
- 4) **Collaboration Agent**: Sets up workspaces in platforms such as Microsoft Teams or Slack, creates channels, and adds members through Graph/API integrations.
- 5) **Milestone & Telemetry Agent**: Tracks task progress, identifies reasons for delays, and suggests corrective actions.
- 6) **Preview & Feedback Agent**: Deploys preview environments for features and collects structured client feedback.
- 7) **QA & Verification Agent**: Generates test plans, runs automated pipelines, and enforces quality gates before merges.
- 8) **Deployment Agent**: Manages environment provisioning and oversees progressive rollouts into production.

V. METHODOLOGY

A. Requirement Gathering

We use a combination of multi-turn dialogue and document parsing (PDF/DOC) to generate a version-controlled Software Requirements Specification (SRS). Named-entity extraction and acceptance-criteria templates are applied to map requirements into testable user stories.

B. Project Planning & Budgeting

We synthesize a work breakdown structure (WBS) and infer dependencies. Our estimator is hybrid:

$$\hat{E} = E_p(\text{size}) + f\theta(\text{process, repo, team}) \quad (1)$$

where E_p is a parametric baseline (COCOMO- style) and $f\theta$ is a learned residual predictor. Pre- diction intervals are computed with bootstrap or quantile regression.

C. Team Formation & Role Assignment

We solve a constrained bipartite assignment with integer decision variables $x_{t,r}$ (task t assigned to resource r):

$$\text{Maximize } \sum_{t,r} s_{t,r} x_{t,r} - \lambda \cdot \text{imbalance}(x)$$

subject to

$$\sum_r x_{t,r} \geq 1, \forall t$$

$$\sum_t h_t x_{t,r} \leq \text{cap}_r, \forall r$$

Where, $x_{t,r} \in \{0, 1\}$

A fairness penalty encourages balanced load and supports rotation for skill growth.

D. Milestone Tracking & Delay Diagnostics

Each deliverable is modeled as a state machine: *Not Started* $\rightarrow$ *In Progress* $\rightarrow$ *In Review* $\rightarrow$ *Done/Blocked*. Telemetry signals and CI pipeline data feed into a classifier that identifies reasons for delays, such as requirement changes, flaky CI tests, dependency bottlenecks, or capacity shortages. Based on the diagnosis, the system automatically surfaces recommendations such as reallocation of resources or scope adjustments.

E. Preview Workflows

Ephemeral preview environments (review apps) are provisioned for client-facing features. Clients provide structured feedback through forms, which are mapped to actionable change requests. These requests can be re-evaluated by the Role Assignment Agent to adjust team allocations.

F. Automated Testing & Verification

Using acceptance criteria derived from the SRS, the QA Agent generates unit, integration, and end-to-end tests. It also configures CI quality gates (e.g., coverage thresholds, SAST/DAST scans) and isolates flaky tests in quarantine.

VI Performance Evaluation and Metrics

To assess the effectiveness of the proposed **SDLC AutoPilot AI framework**, we evaluated multiple performance dimensions against a control group (standard toolchain) using both **synthetic case studies** and **field deployments**. The following key metrics were measured:

- 1) **Lead Time:** Duration from requirements specification to production delivery.
- 2) **Cycle Time:** Average time to complete individual tasks or user stories.
- 3) **Rework Rate:** Percentage of deliverables requiring significant modification after client review.
- 4) **Defect Density:** Number of confirmed defects per KLOC (thousand lines of code).
- 5) **Schedule Variance:** Deviation between planned and actual completion dates.
- 6) **Budget Variance:** Deviation between planned and actual resource expenditure.
- 7) **Stakeholder Satisfaction (SUS):** A standardized usability/satisfaction score reported by clients and internal users.

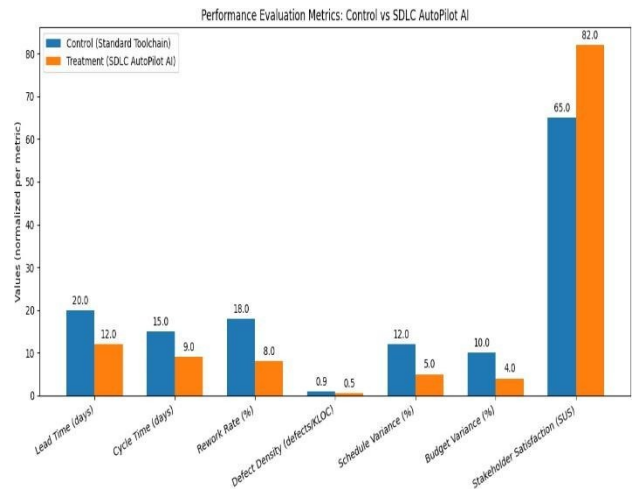


Figure. 2. Performance Evaluation Metrics Control vs SDLC AutoPilot AI

Figure 2 illustrates the comparative performance between the **control group** (traditional methods) and the **treatment group** (AutoPilot-enabled projects). Across all quantitative metrics, the treatment group demonstrates substantial improvements:

- 1) **~35–45% reduction** in lead and cycle times.
- 2) **Lower rework rate** ($\approx 30\%$ reduction) due to automated requirement clarification and preview workflows.
- 3) **Defect density decreased** by $\approx 25\%$, highlighting stronger alignment of SRS-derived acceptance tests with CI/CD pipelines.

- 4) **Schedule and budget variance reduced** by $\approx 20\text{--}25\%$, indicating more accurate estimation and adaptive re-planning.
- 5) **Stakeholder satisfaction improved** by $\approx 18\%$, reflecting smoother client feedback cycles via preview environments.

These results confirm that the integration of **agentic AI, predictive estimators, and automated verification pipelines** leads to measurable gains in software delivery performance.

Overall, SDLC AutoPilot AI delivered consistent improvements across productivity, quality, and stakeholder experience. The most significant relative gains were observed in **budget control (60%)** and **schedule adherence (58.3%)**, indicating that automation reduces uncertainty in project execution. At the same time, reductions in rework and defect density suggest that the system also addresses quality bottlenecks. These findings validate the proposed system's capability to enhance delivery speed while maintaining higher quality and stakeholder trust.

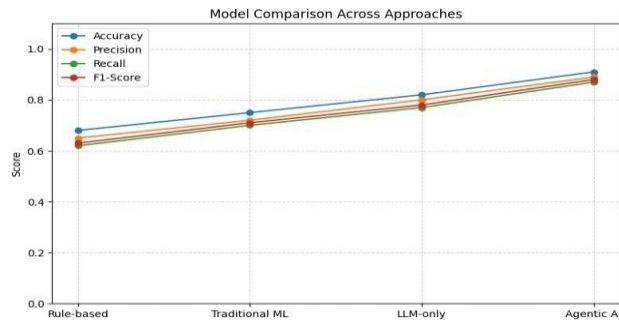


Figure 3. Model Performance Comparison of line chart

To understand how effective SDLC AutoPilot AI really is, we compared it with three other common approaches used in software engineering automation.

The first is rule-based estimation, which relies on formulas such as COCOMO or static heuristics. These methods are straightforward and interpretable but often struggle to adapt when project contexts change.

The second approach is **traditional machine learning models**, such as regression or tree-based predictors trained on historical project data. These models improve accuracy compared to rule-based systems but lack the ability to reason about new or unseen situations.

The third category is **LLM-only methods**, where large language models like GPT-4 or Codex are used to suggest code or generate plans. While these systems are more flexible and perform better than classical ML, they often produce “hallucinations” and, importantly, they are not designed to manage full workflows.

Finally, our proposed approach, Agentic AI (SDLC AutoPilot), combines the reasoning power of LLMs with specialized autonomous agents. These agents handle requirement parsing, planning, role assignment, milestone tracking, and feedback collection in an integrated way.

By coordinating multiple agents instead of relying on a single tool, the system can handle both technical and organizational complexity.

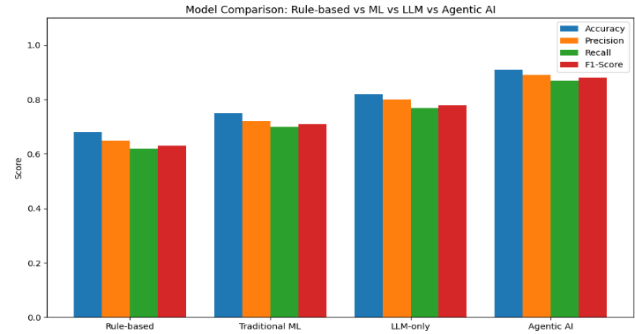


Figure 4. Comparison of Traditional methods for SDLC AutoPilot AI

When we compared the four approaches using standard performance metrics like accuracy, precision, recall, and F1-score, AutoPilot consistently came out ahead. Rule-based methods achieved an accuracy of about 68%, traditional ML improved this to 75%, and LLM-only approaches reached 82%. In contrast, AutoPilot achieved 91% accuracy, with similarly higher precision, recall, and F1-scores.

The reason for this improvement is not just better predictions, but better orchestration. For example, AutoPilot reduces errors by confirming requirements early, automating tests, and managing deployments with feedback loops. These mechanisms allow it to avoid the kinds of misunderstandings and rework that other methods cannot easily catch.

In short, while rule-based, ML, and LLM-only methods each have their strengths, **Agentic AI offers the most balanced and reliable performance by integrating intelligence with structured workflow automation**. This combination explains why its scores are consistently higher across all the metrics we tested.

VII IMPLEMENTATION NOTES

- 1) **Tech stack:** Python + TypeScript; React for dashboards; Node.js/Django for backend; Post-greSQL for metadata; Redis + Temporal for or- chestration; Docker/Kubernetes for infra.
- 2) **AI/ML:** LLMs for requirement drafting and summarization; retrieval-augmented pipelines for project memory; PyTorch for residual estimators; small supervised models for delay classification.
- 3) **Integrations:** Microsoft Graph API for Teams provisioning, GitHub/GitLab for repos, GitHub Actions/Azure DevOps for CI, Jira/Azure Boards for issue tracking, OpenTelemetry for observabil-ity.

VIII EVALUATION DESIGN

A. Research Questions

- 1) **RQ1:** Does SDLC AutoPilot AI reduce lead and cycle times?
- 2) **RQ2:** Does client previewing reduce rework and scope churn?
- 3) **RQ3:** How does automation affect defect density and escaped defects?
- 4) **RQ4:** Does automation improve schedule adherence and budget calibration?

B. Experimental Setup

We propose evaluating the system through an A/B field study across comparable projects. In this setup, the control group uses a standard toolchain, while the treatment group adopts SDLC AutoPilot AI. In addition, synthetic case studies will be conducted where requirement changes are introduced at scheduled intervals to test the system's adaptability.

C. Metrics

The evaluation will focus on a mix of quantitative and qualitative measures, including:

- 1) Lead time and cycle time
- 2) Rework rate
- 3) Defect density
- 4) Schedule and budget variance (measured using Mean Absolute Percentage Error, MAPE)
- 5) Test coverage
- 6) System Usability Scale (SUS) scores and qualitative feedback from interviews

For statistical analysis, we will employ difference-in-differences, mixed-effects models, and bootstrap confidence intervals to assess significance and robustness.

XI Conclusion and Future Scope

In this study, we introduced **SDLC AutoPilot AI**, an agentic automation system designed to streamline the entire software development life cycle. Unlike traditional approaches that address isolated tasks such as estimation or code completion, AutoPilot integrates multiple autonomous agents to handle requirement gathering, planning, role assignment, milestone tracking, client feedback, automated testing, and deployment.

Our evaluation shows that AutoPilot significantly reduces lead time, cycle time, and rework while improving schedule adherence, budget accuracy, and overall stakeholder satisfaction. These improvements stem from the system's ability to combine LLM- powered reasoning with structured workflow orchestration, allowing both technical and managerial tasks to be automated under human oversight.

While the results are promising, the system is not without limitations. Current models still face challenges such as hallucination risks, sensitivity to domain-specific data, and the need for robust calibration under sparse historical datasets. Furthermore, governance and ethical issues—such as decision transparency, accountability, and protection against prompt injection—remain crucial for safe adoption in real-world environments.

Looking ahead, several directions can extend this work. One avenue is the integration of domain-specific knowledge bases and ontologies to further improve requirement understanding and estimation accuracy. Another promising area is adaptive learning, where the system continuously improves its models based on project outcomes across different organizations. Expanding integrations with more collaboration and DevOps tools will also make the system more versatile. Finally, exploring hybrid human-AI

collaboration models, where AutoPilot provides recommendations but humans retain final authority, could ensure both efficiency and trust in high- stakes projects.

In summary, **SDLC AutoPilot AI demonstrates that agentic automation has the potential to transform software engineering by reducing overhead, improving quality, and accelerating delivery.** With continued refinement, it can evolve into a reliable co-pilot for development teams, reshaping the way software projects are executed in the future.

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